MIL-O-82246D
21 January 1982
SUPERSEDING
MIL-O-82246C
24 April 1975

MILITARY SPECIFICATION

OVERSHOES, RADIOACTIVE CONTAMINANTS, PROTECTIVE

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers the requirements for a low unsupported molded overshoe made of natural or synthetic rubber or a combination of both.

1.2 Classification. The overshoes covered by this specification shall be of one type and in the designs specified herein.

1.2.1 Sizes. The overshoes shall be in the following sizes to accommodate the related size of shoe (see 6.2).

Size	Related shoe size
Small	6 to 7-1/2
Medium	8 to 9
Large	9-1/2 to 11
Extra Large	11-1/2 to 13

2. APPLICABLE DOCUMENTS

2.1 Issue of documents. The following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein:

Beneficial comments (recommendations, additions, and deletions) and any pertinent data which may be of use in improving this document should be addressed to: Officer in Charge, Navy Clothing and Textile Research Facility, 21 Strathmore Road, Natick, MA 01760 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

FSC 8430

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SPECIFICATION

MILITARY

MIL-B-17757 - Boxes, Shipping, Fiberboard, (Modular Sizes)

STANDARDS

FEDERAL

FED-STD-595 - Colors
FED-STD-601 - Rubber, Sampling and Testing

MILITARY

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-284 - Visual Inspection Guide for Rubber Footwear

(Copies of specifications, standards, drawings, and publications required by contractor in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer. Requests for individual color chips (3 x 5 supplements) should be submitted to the Naval Publications and Forms Center, 5081 Tabor Ave, Philadelphia, PA 19120, citing FED-STD-595 and appropriate chip number.)

LAWS AND REGULATIONS

U.S. POSTAL SERVICE MANUAL

(Copies of the manual may be obtained from the Superintendent of Documents, U.S. Government Printing Office, Washington, D.C. 20402.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.

AMERICAN SOCIETY FOR TESTING AND MATERIALS STANDARDS

Method ASTM D-1630- Abrasion Resistance of Rubber Soles and Heels

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103.)

NATIONAL MOTOR FREIGHT TRAFFIC ASSOCIATION, INC., AGENT
National Motor Freight Classification

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(Application for copies should be addressed to American Trucking Association, ATTN: Traffic Department, 1616 P Street, N.W., Washington, D.C. 20035.)

UNIFORM CLASSIFICATION COMMITTEE, AGENT
Uniform Freight Classification

(Application for copies should be addressed to the Uniform Classification Committee, Room 1106, 222 South Riverside Plaza, Chicago, IL 60606.)

3. REQUIREMENTS

3.1 Guide Samples. Samples, when furnished, are solely for guidance and information to the contractor (see 6.3). Variation from this specification may appear in the sample, in which case this specification shall govern.

3.2 First article. When specified (see 6.2) the contractor shall furnish sample unit(s) for first article inspection and approval (see 4.3).

3.3 Materials

3.3.1 Rubber compound. The compound shall be made from virgin natural rubber, synthetic rubber or a combination of both. The compound shall meet the physical properties specified in Table I when tested as specified in 4.1.1.

Table I - Physical requirements

Characteristic	Requirement
Tensile strength:	
Initial	2000 psi (minimum)
Loss after accelerated aging	30 percent (maximum)
Tensile stress:	
200 percent elongation	200 psi (minimum)
400 percent elongation	650 psi (minimum)
600 percent elongation	1500 psi (minimum)
Elongation at rupture:	
Initial	700 percent (minimum)
Loss after accelerated aging	30 percent (maximum)
Hardness (durometer)	50 (± 5)
Abrasive index:	
Initial	35 (minimum)
After accelerated aging	25 (minimum)

3.4 Color. The color of the rubber compound forming the vamp, sole, instep, and heel shall be black. There shall be two yellow identification patch areas on each overshoe, one in the toe area covering 2 to 5 square inches and the other on the rear quarter covering 15 to 20 square inches (see Figure 1). The yellow shall approximate Shade 13655 of FED-STD-595.

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3.5 Thickness. The thickness of the sole and heel excluding tread shall be a minimum of 0.100 inch. The thickness of the sole and heel including tread shall be a minimum of 0.135 inch. The minimum thickness of the upper shall be 0.030 inch except the reinforced portions shall be a minimum of 0.0375 inch.

3.6 Finish. The overshoe shall be cured and treated to eliminate tackiness and have a smooth finish. All mold flashings shall be trimmed and all inside corners rounded and made smooth. The yellow rubber compound for the patched area shall be molded simultaneously with the overshoe.

3.7 Design. The overshoes shall be constructed as an unlined unit on the contractor's mold to provide proper fit for the related sizes and shapes of men's work shoes. The overshoes may be of a "storm" or "sandal" design as shown in Figure 1. They shall consist of an upper, sole, and heel. The tread pattern shall be a continuous grid type and shall not be subject to loss of pieces in wear. The tread shall provide good traction.

3.7.1 Figure. Figure 1 is furnished for information purposes only. To the extent of any inconsistencies between the written specification and the figure, the written specification shall control.

3.8 Marking. The size and contractor's name shall be molded on the outsole in the shank area with characters not less than 1/4 inch high. "Left" and "right" designation may be applied.

3.9 Workmanship. The finished overshoes shall conform to the quality established by this specification. Occurrence of defects shall not exceed the applicable acceptable quality levels.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in this specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Certificate of compliance. Where certificates of compliance are submitted, the Government reserves the right to check test such items to determine the validity of the certification.

4.2 Classification of inspection. The inspection requirements specified herein are classified as follows:

1. First article inspection (see 4.3).
2. Quality conformance inspection (see 4.4).

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4.3 First article inspection. When required, the first article, submitted in accordance with 3.2, shall be inspected as specified in 4.4.2.1 and 4.4.2.2 for compliance with design, construction, workmanship and dimensional requirements.

4.4 Quality conformance inspection. Sampling for inspection shall be performed in accordance with MIL-STD-105, except where otherwise indicated herein or in MIL-STD-284.

4.4.1 Component and material inspection. In accordance with 4.1 above, components and materials shall be inspected and tested in accordance with all requirements of referenced specifications, drawings, and standards unless otherwise excluded, amended, modified, or qualified in this specification or applicable purchase document.

4.4.1.1 Testing of rubber compound. Unless otherwise specified, the tests shown in Table II shall be conducted in accordance with FED-STD-601. The physical values specified in Section 3 shall be applicable to the lot average. A lot shall consist of the amount of rubber compounded in a day. The sample unit for testing shall be a representative composite taken from a lot. Two specimens 6 by 6 inches shall be prepared from the composite for each of the prescribed tests. The specimens for the abrasion index and hardness tests shall be 0.250 inch thick and 0.075 inch thick for the other tests. The lot shall be unacceptable if two units fail any of the prescribed tests.

Table II - Tests

Characteristics	Reference	
	Requirement paragraph	Test method
Tensile stress: 1/		
200 percent	3.3.1	4131
400 percent	3.3.1	4131
600 percent	3.3.1	4131
Tensile strength:		
Initial	3.3.1	4111
After aging	3.3.1	4111, 7221
Elongation:		
Initial	3.3.1	4121
After aging	3.3.1	4121, 7221
Hardness	3.3.1	3021
Abrasive index:		
Initial	3.3.1	ASTM Method D-1630
After aging	3.3.1	ASTM Method D-1630 7221

1/ The modulus curve shall be carried to the break point.

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4.4.2 Examination of the end item. Examination of the end item shall be in accordance with 4.4.2.1 through 4.4.2.3.

4.4.2.1 Visual examination. Defects shall be classified in accordance with the classification of defects shown in Table I of MIL-STD-284 and in Table III of this specification. The lot size shall be expressed in units of one overshoe. The sample unit shall be one overshoe and the selection shall be by pairs. For pairing examination, the pair shall be examined together.

Table III - Classification of defects

Defects	Classification Major	Minor
Pairing		
Not mated; not a right and a left of an identical size	X	
Design and construction not as specified	X	
Identification color patch missing	X	
Finish not smooth or poor or uneven molding	X	
Not of uniform color		X
Tread pattern not continuous grid type	X	
Incomplete trimming of flashing		X
Marking missing, incorrect, illegible, or not located as specified	X	
Any evidence of insecure bonding	X	

4.4.2.2 Thickness examination. The overshoes shall be examined for conformance to the minimum values specified in 3.5, and shall be measured using any suitable measuring device accurate to 0.0005 inch. Any measurement less than the specified minimum shall be classified as a defect. The lot size shall be expressed in units of one overshoe. The sample unit shall be one overshoe.

4.4.2.3 Acceptable quality levels and inspection levels. The acceptable quality levels (AQL's) and inspection levels shall be in accordance with the following listing. AQL's are expressed in defects per hundred units.

	AQL's	Inspection level
For defects listed in 4.4.2.1:		
Major defects	6.5	II
Total defects (Major and Minor combined)	15.0	II
For defects listed in 4.4.2.2:	4.0	S-3
For defects listed in 4.4.3:	2.5	S-2

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4.4.3 Examination of packaging requirements. An examination shall be made to determine that preservation packaging, packing, and marking comply with Section 5 requirements. Defects shall be scored in accordance with the list below. The sample unit shall be one shipping container fully prepared for delivery except that it need not be sealed. Defects of closure shall be examined on shipping containers fully prepared for delivery. The lot size shall be the number of shipping containers in the end item inspection lot.

Examine	Defect
Marking (exterior)	Omitted, incorrect, illegible, of improper size, location, sequence or method of application.
Materials	Any component missing or not as specified. Any component damaged.
Workmanship	Inadequate application of components, such as: Incomplete closure of container flaps, loose strapping, inadequate stapling. Bulging or distortion of containers.
Content	Number of items per container is more or less than specified.

5. PACKAGING

5.1 Preservation Packaging. Packaging shall be level A or C (commercial) as specified (see 6.2).

5.1.1 Level A. Each pair of mated overshoes of one size only, shall be placed heel to toe with the uppers facing each other. Each pair of overshoes shall be bound with an elastic band, or cotton tape or twine.

5.1.2 Level C (Commercial). Overshoes shall be packaged to afford adequate protection against deterioration and physical damage during shipment from the supply source to the first receiving activity. The package and the quantity per package shall be the same as that normally used by the contractor for retail distribution.

5.2 Packing. Packing shall be level A, B, or C (Commercial) as specified (see 6.2).

5.2.1 Level A. Sixty pairs of overshoes, packaged as specified in 5.1, shall be packed in a fiberboard shipping container conforming to class weather-resistant, grade V2s, size 3A of MIL-B-17757, assembled, closed and reinforced in accordance with the requirements of the appendix of that specification and referenced documents.

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5.2.2 Level B. Sixty pairs of overshoes, packaged as specified in 5.1, shall be packed in a fiberboard shipping container conforming to class domestic, grade 275, size 3A of MIL-B-17757, assembled and closed according to the requirements of the appendix of that specification and referenced documents.

5.2.2.1 Weather-resistant fiberboard container. When specified (see 6.2), the shipping container shall be a grade V3c, W5c, or W6c fiberboard box fabricated in accordance with MIL-B-17757 and closed in accordance with the appendix of the box specification.

5.2.3 Level C (Commercial). Overshoes, packaged as specified in 5.1, shall be packed in a manner to insure carrier acceptance and safe delivery at destination at the lowest transportation rate for such supplies. The quantity per shipping container shall be the same as that normally used by the contractor for retail distribution. Containers shall comply with the U.S. Postal Service Manual, Uniform Freight Classification Rules or National Motor Freight Classification Rules, as applicable.

5.3 Marking. Shipping containers shall be marked in accordance with Standard Industry practices and shall include the following information:

Federal Stock Number
Quantity, Unit and Size
Contractor's Name
Contract Number

5.3.1 Mixed sizes. Each shipping container packed with mixed sizes shall have securely attached directly under the container marking, a plain white paper label 5 by 4 inches, with the inscription "MIXED SIZES" and the correct quantity of sizes contained within.

6. NOTES

6.1 Intended use. The overshoes are intended for protection of personnel in the presence of low level radioactive contaminants.

6.2 Ordering data. Procurement documents should specify the following:

(a) Title, number and date of this specification.
(b) Sizes required (see 1.2).
(c) Whether first article sample is required (see 3.2).
(d) Levels of packaging and packing required (see 5.1 and 5.2).
(e) When weather-resistant grade fiberboard shipping containers are required for level B packing (see 5.2.2.1).

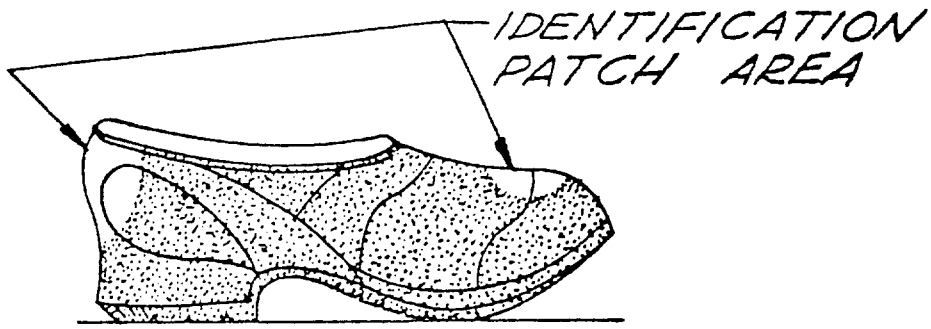
6.3 Samples. For access to samples, address the procuring activity issuing the invitation for bids.

Custodians:
Navy - NU
Air Force - 11
Review activities:
Navy - SH
Air Force - 99
DLA - CT
User Activities:
Navy - AS, OS

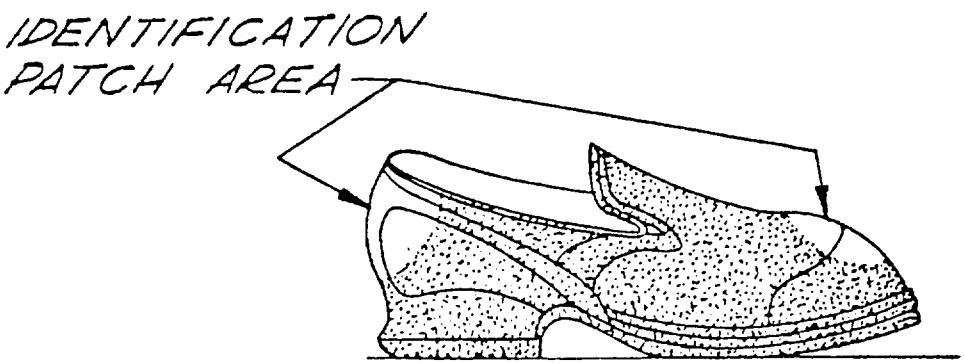
Preparing activity:
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SANDAL DESIGN



STORM DESIGN

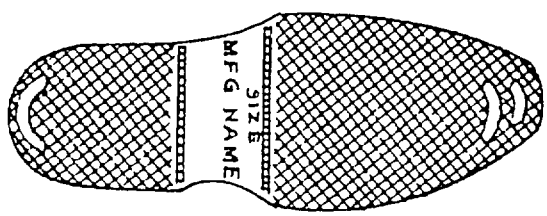


FIG. 1. OVERSHOES, RADIOACTIVE CONTAMINANTS, PROTECTIVE

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NAME OF ORGANIZATION AND ADDRESS OF SUBMITTER

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